

1 Watershed-scale PFAS fate and transport across surface
2 water and groundwater: a two-way coupled
3 SWAT⁺/MODFLOW 6 model quantifies the legacy
4 groundwater contribution to in-stream loads

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6 **Highlights**

- 7 • PFAS fate-and-transport ported into SWAT⁺ (three-phase soil sorption)
- 8 • First two-way SWAT⁺–MODFLOW 6 coupling: PFAS to groundwater and back
- 9 • Dual-compartment surface-water and groundwater PFAS mass balance, public-data
10 build
- 11 • Streambed porewater fingerprint validates the groundwater PFAS discharge pathway
- 12 • Systematic global sensitivity and uncertainty analysis of the coupled model

13 **Abstract**

14 At watersheds with a legacy source, per- and polyfluoroalkyl substances (PFAS)
15 reach streams not only over the land surface but through groundwater discharge —

yet an in-stream sample cannot separate the two, and watershed models have carried PFAS only from the surface into the subsurface, leaving the groundwater contribution to in-stream PFAS, the quantity source attribution depends on, unquantified. We advance watershed-scale PFAS fate and transport by resolving the two-way exchange between surface water and groundwater: a process-based three-phase soil model (aqueous, Freundlich solid-phase, and air–water interfacial sorption) carries PFAS from the land to the channel, and a coupled MODFLOW 6 groundwater flow-and-transport model returns the groundwater-borne PFAS to the stream in one continuous mass balance. On the Rogue River, Michigan, downstream of the Wolverine World Wide legacy source, the model is validated against an unusually rich record — 5,383 water-table heads (Nash–Sutcliffe efficiency 0.91), hundreds of groundwater PFOS measurements, streambed pore water at the discharge interface, and a multi-analyte in-stream network. Two independent lines of evidence — a model-free multi-analyte fingerprint decomposition and a joint surface/groundwater calibration that avoids double-counting — show that legacy groundwater discharge is a first-order, spatially localized contributor to the lower-mainstem in-stream PFOS load. A systematic global sensitivity and uncertainty analysis identifies the controls on how PFAS partitions between the surface and groundwater pathways. The result is a pathway-resolved, mass-conserving basis for attributing in-stream PFAS to its sources.

Keywords: SWAT⁺; MODFLOW 6; surface-water/groundwater coupling; PFAS; groundwater transport; streamflow routing (SFR/SFT); pilot-point calibration

1 Introduction

Per- and polyfluoroalkyl substances (PFAS) are persistent, mobile synthetic chemicals now regulated in drinking water at the nanogram-per-litre level (Buck et al., 2011; Guelfo et al., 2021; U.S. Environmental Protection Agency, 2024c). The PFAS measured in an in-stream grab sample is the sum of several transport pathways acting at once — surface runoff, lateral

and sediment-bound transport, biosolids and industrial inputs, and the discharge of contaminated groundwater (Prevedouros et al., 2006; Sepulvado et al., 2011). Because PFAS management now turns on *where* the in-stream contamination originates — a Clean Water Act load allocation, a CERCLA cost recovery, a source-control or remediation decision (U.S. Environmental Protection Agency, 2024b; Michigan Department of Environment, Great Lakes, and Energy (EGLE), 2022) — the governing scientific need is no longer to detect PFAS but to attribute an observed in-stream concentration to the pathways that produced it.

Among those pathways, groundwater discharge is at once the most consequential and the least resolvable. Where a legacy source exists, PFAS that infiltrated decades ago migrates through the aquifer and re-enters the stream as baseflow, so that a substantial share of the in-stream load can arrive through the streambed rather than across the land surface (Rafiei and Nejadhashemi, 2023). Yet this is exactly the pathway that observation cannot disaggregate: a grab sample cannot separate groundwater-borne from surface-derived PFAS, and the statistical occurrence models that dominate large-scale PFAS prediction return a probability of contamination, not a mass-conserving, pathway-resolved concentration (Tokranov et al., 2024). Quantifying the groundwater contribution to in-stream PFAS therefore requires a process model that carries the contaminant continuously from the land surface, through the aquifer, and back to the channel.

Watershed-scale PFAS modeling has been advancing toward this goal. PFAS was first simulated in receiving rivers on a hydrology supplied by a coupled surface-water/groundwater model (Raschke et al., 2022), and a distributed watershed model subsequently represented PFAS fate and transport across the land surface and identified groundwater as a watershed-scale source (Rafiei and Nejadhashemi, 2023). These advances established that process-based watershed PFAS modeling is tractable and that the subsurface is material. They share one structural limit: the contaminant is passed in a single direction, from the surface model into the subsurface, so the PFAS that a legacy plume discharges back to the stream is never returned to the surface-water mass balance. The groundwater *contribution* to the in-stream

load — the very quantity that source attribution depends on — thus remains outside the model.

We advance watershed-scale PFAS fate and transport by resolving this missing direction: the two-way exchange of PFAS between surface water and groundwater. A process-based, three-phase soil model carries PFAS from the soil profile to the stream, partitioning among the aqueous phase, the Freundlich solid phase (Higgins and Luthy, 2006), and the air–water interface that governs retention in unsaturated media (Costanza et al., 2019; Brusseau, 2018; Guo et al., 2020); a groundwater flow-and-transport model then carries the infiltrated PFAS through the aquifer and returns the groundwater-borne fraction to the channel, closing a single continuous PFAS mass balance across the two compartments. The subsurface leg is built on MODFLOW 6, whose native Freundlich groundwater transport and advanced streamflow transport (Langevin et al., 2017; Morway et al., 2021) provide the contaminant physics that the embedded-solver couplings underlying earlier watershed models do not carry (Bailey et al., 2016, 2025); fully integrated hydrologic simulators exist outside the watershed-modeling family but do not carry a watershed’s surface PFAS budget (Markstrom et al., 2008; Brunner and Simmons, 2012).

This capability lets us pose a question the one-directional models could not answer: at an impacted watershed, how much of the in-stream PFAS is groundwater-borne, and where along the river does it enter? We address it on the Rogue River, Michigan, downstream of the Wolverine World Wide / House Street legacy PFAS source — a tannery-waste disposal area on a groundwater divide immediately upgradient of the river, where source-area groundwater has reached the order of 10^5 ng L⁻¹ and where a groundwater interceptor was later engineered specifically to capture PFAS before it discharges to the channel (U.S. Environmental Protection Agency, Region 5, 2024; Michigan Department of Environment, Great Lakes, and Energy (EGLE), 2025, 2021). The site offers an unusually rich public record — thousands of water-table observations, hundreds of groundwater PFAS measurements, streambed pore water sampled at the discharge interface, and a multi-analyte in-stream network (Michigan

Department of Environment, Great Lakes, and Energy (EGLE), 2019; Lemke et al., 2024)
— against which every compartment of the model can be tested (study-area configuration
in Table 1).

Two independent lines of evidence then converge on the answer. A model-free, multi-analyte fingerprint decomposition of the in-stream signal shows that the rising PFOS gradient along the lower mainstem carries the chemical signature of the groundwater plume; and a joint surface/groundwater calibration — constructed to avoid the double-counting that a naive sum of the two legs would incur — attributes a first-order, spatially localized share of the lower-mainstem load to legacy groundwater discharge. We further report the first systematic global sensitivity and uncertainty analysis of coupled watershed PFAS fate and transport, isolating the parameters that govern how PFAS is partitioned between the surface and subsurface routes. The result is a pathway-resolved, mass-conserving account of PFAS fate and transport across the surface-water and groundwater compartments of a single watershed — the basis for attributing in-stream PFAS to its sources and for evaluating the pathway-specific actions that PFAS management requires.

2 Methods

2.1 The SWAT⁺ PFAS surface-water engine

2.1.1 Design principle

The engine is implemented in modern, free-form Fortran SWAT⁺ as a constituent that rides the existing spatial-object and constituent-hydrograph infrastructure. Its design separates the two halves of the problem along the line of where PFAS behavior is genuinely distinctive. The soil column—where PFAS partition simultaneously among water, solids, and the air–water interface—implements the process-based three-phase model of Rafiei and Nejadhashemi (2023), ported from legacy SWAT (where it was first developed within a SWAT–

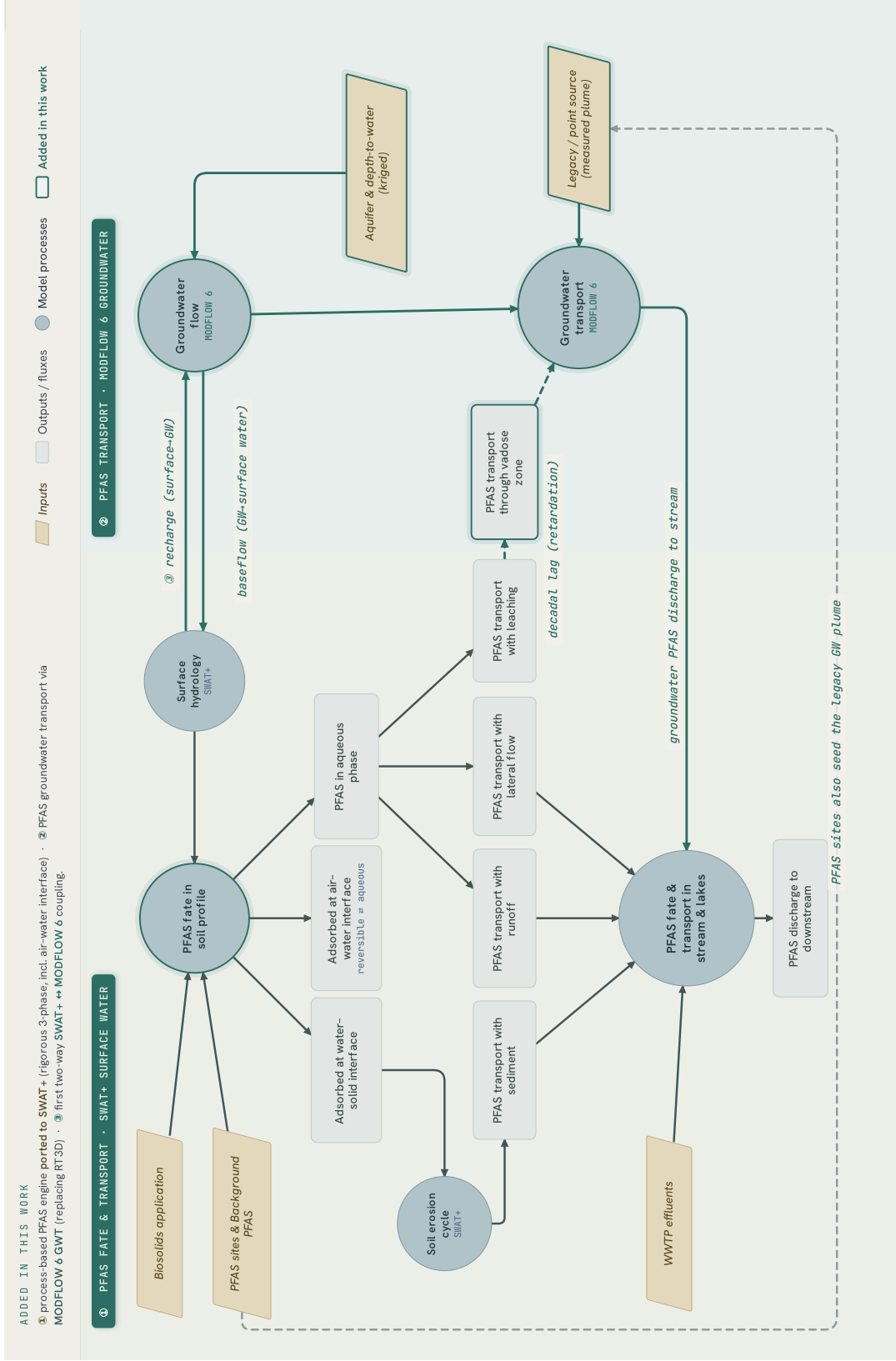


Figure 1: Conceptual model of the coupled SWAT⁺/MODFLOW 6 PFAS fate-and-transport framework. The surface-water leg partitions PFAS in the soil profile across the aqueous, solid, and air-water-interface phases and routes it to streams with runoff, lateral flow, and sediment. New in this work (highlighted): a process-based PFAS engine ported to SWAT⁺; PFAS groundwater transport in MODFLOW 6 (replacing RT3D); and the first two-way SWAT⁺-MODFLOW 6 coupling — the daily recharge/baseflow water exchange, the retardation-lagged vadose pathway, the legacy groundwater plume, and the groundwater PFAS discharge to streams.

MODFLOW–RT3D framework) into the modern SWAT⁺ engine. The channel and reservoir network—where PFAS behave, to first order, as a strongly sorbing conservative tracer with settling and benthic exchange—reuses the mathematics of the established SWAT⁺ pesticide subsystem, restricted to its linear-partition subset. This separation makes the engine both economical (most of the routing machinery already exists and is tested) and auditable (the ported code is confined to a small, independently verifiable set of routines).

2.1.2 Soil three-phase equilibrium

For each hydrologic response unit (HRU) and soil layer, the total PFAS mass present at the start of a day is partitioned among three coexisting phases at local equilibrium: a dissolved aqueous phase, a solid-sorbed phase described by a Freundlich isotherm, and an interfacially adsorbed phase at the air–water interface described by a Langmuir isotherm, the air–water interfacial term being the mechanism that can dominate PFAS retention in unsaturated media (Brusseau, 2018; Guo et al., 2020; Rafiei and Nejadhashemi, 2023). Writing C_w for the aqueous concentration, the layer mass balance is

$$M = \theta V C_w + m_s k_f C_w^n + \Gamma_{\max} \frac{K_L C_w}{1 + K_L C_w} A_{aw}, \quad (1)$$

where M is the layer PFAS mass, θ the volumetric water content, V the layer volume, m_s the solid mass, (k_f, n) the Freundlich coefficient and exponent, $(\Gamma_{\max}, K_L)$ the Langmuir maximum surface excess and affinity, and A_{aw} the air–water interfacial area, estimated from soil water content and median grain diameter. Because the right-hand side of Eq. (1) is strictly monotonic increasing in $C_w > 0$, the equation has a unique positive root, which the engine isolates by a bracketed bisection and accelerates with an Improved-Halley iteration, retaining the bisection bracket as a guaranteed fallback. The original SWAT-MODFLOW-RT3D implementation solved this balance in quadruple precision (Rafiei and Nejadhashemi, 2023); we re-implemented it in double precision and verified that the choice is safe: against

a 40-digit decimal reference, the double-precision solver holds a worst-case relative error of 1.7×10^{-7} across 29,160 cases spanning the stiff corners of the parameter space (Freundlich n down to 0.31, layer mass down to 10^{-6} kg ha⁻¹, and dry-to-saturated water contents), with no non-convergence. Once C_w is known, the dissolved, solid, and interfacial masses follow by back-substitution and set the concentrations available to each loss pathway.

2.1.3 Land-phase mobilization

Following the equilibrium solve, dissolved PFAS is removed from the surface layer in proportion to surface runoff and percolation through a runoff/leaching partition coefficient, from deeper layers in proportion to lateral subsurface flow, and the solid-sorbed fraction is removed in proportion to sediment yield by an enrichment-ratio formulation, exactly paralleling the SWAT⁺ pesticide land phase so that the soil pool decrements consistently across all four pathways. The four daily losses—surface runoff, lateral flow, leaching, and sediment-bound—are accumulated per HRU and per PFAS compound and become the constituent loads delivered to the channel network. A whole-run land-phase mass balance confirms closure: for every HRU the change in soil pool equals the sum of the four pathway losses to a relative residual of order 10^{-5} .

2.1.4 In-stream transport and cross-validation

In the channel, PFAS partitions between a dissolved and a sediment-sorbed fraction by a linear organic-carbon coefficient, $f_{\text{sol}} = 1/(1 + k_d c_s)$, where k_d is the partition coefficient scaled by reach bed-carbon fraction and c_s the suspended-sediment concentration, following the organic-carbon normalization of sediment–water partitioning for hydrophobic pollutants (Karickhoff et al., 1979; Chapra, 1997). The sorbed fraction settles to the bed; bed PFAS resuspends, diffuses across the sediment–water interface, and is permanently lost by burial; the dissolved and sorbed fractions leave the reach in the outflow. These are precisely the non-reactive terms of the SWAT⁺ pesticide channel solver, with the pesticide-specific aque-

ous decay, volatilization, and metabolite production removed because PFOS is non-volatile and recalcitrant on the relevant timescale. We exploit this correspondence as a correctness check. Configuring a pesticide with unit decay factors, zero volatilization, and no metabolites reduces the pesticide solver exactly to the PFAS linear-partition subset; driving both routines with identical inflow and parameters over a 60-day synthetic reach sweep, including a dry-reach day, the new PFAS routine and the established pesticide routine agree on all nine shared mass terms to a maximum absolute difference of 2.4×10^{-7} kg and a maximum relative difference of 2.6×10^{-7} —single-precision floating-point tolerance. The in-stream mathematics is therefore not novel and not a source of error; the novelty is the soil column that supplies its loads.

2.1.5 Reservoirs and point sources

Reservoirs are treated as mass-conserving completely mixed cells: the PFAS entering a reservoir on a given day is added to the stored mass and the outflow carries the fraction leaving with the day’s discharge, with the remainder retained, so that no mass is created or destroyed at impoundments (a benthic burial sink can be enabled but is omitted here, PFOS being mobile). Point sources—contaminated-site leachate, wastewater effluent, or firefighting-foam discharge—are injected as a per-reach daily load read from an optional input file and added to the reach inflow before routing; the injected mass is tracked in the in-stream mass balance. Because routing PFAS through a real network requires the constituent-hydrograph infrastructure that SWAT⁺ allocates only when a constituent is declared, the engine activates that infrastructure itself when PFAS is present, so a PFAS simulation requires no dummy constituent. A whole-network in-stream balance closes: cumulative reach inflow minus outflow equals the sum of permanent burial and the change in channel storage to floating-point tolerance, and the point-source input line reconciles exactly with the prescribed load. Daily per-reach output reports the PFAS mass balance and the outflow concentration in ng L⁻¹, the quantity compared against ambient observations. This process-based surface-water engine

supplies the land-derived in-stream PFAS load; the groundwater pathway developed below adds the legacy-plume and baseflow-discharge contribution to the same reaches.

2.2 Study area: the Rogue River and the Wolverine source

We demonstrate the integrated model on the Rogue River watershed (Michigan; USGS gauge 04118500), a 606 km² glacial-drift tributary of the Grand River gauged at Rockford. The watershed contains the Wolverine World Wide tannery legacy and the House Street disposal site, one of the most intensively investigated PFAS contaminated-site corridors in the United States, with a dense ambient surface-water network and — decisively for this work — a dense groundwater PFAS plume dataset. The same SWAT⁺ model used for the surface-water engine (Section 2.1; 17,771 HRUs, 590 channels, generated from NHDPlus High Resolution hydrography) supplies the land surface, recharge, and channel network; the MODFLOW 6 model uses a uniform 250 m grid with three layers ($\approx$ 24,900 active cells), summarized with the rest of the coupled-model configuration in Table 1. The contribution is the *automated, reproducible* construction of the coupled model and its contaminant extension on the SWATGenX platform (Rafiei, 2026b), whose high-resolution SWAT⁺ models run on an accelerated, byte-reproducible engine (Rafiei, 2026a); the same code runs for any U.S. gauge.

2.3 On-the-fly MODFLOW 6 model generation

The complete groundwater model — land surface, layered aquifer geometry, hydraulic properties, recharge, and boundary conditions — is generated from public data at model-build time. Building each component on the fly, rather than from pre-computed raster products, keeps every input auditable and guarantees consistency with the surface-water model. The land surface is the SWAT⁺ DEM; aquifer geometry and conductivity are kriged from a raw well database; and recharge is taken directly from the SWAT⁺ water balance. A uniform-thickness bedrock buffer beneath the drift and water table is required for steady-

state Newton stability on dissected basins, where a sliver bedrock layer in deep-water-table uplands otherwise dewateres and oscillates.

Land surface. The model top is the SWAT⁺ DEM resampled onto the MODFLOW grid centres, the single source of truth for elevation with all other layers snapped beneath it. Sharing the surface-water model’s own elevation eliminates a class of elevation errors that arise when a separate DEM product is clipped and cleaned independently and removes an external dependency.

Aquifer geometry and hydraulic properties. Layer bottoms and hydraulic properties are kriged on the fly from a raw groundwater-well database (in the demonstration region, the Michigan Wellogig database, ≈ 1.17 million wells; a companion effort harmonizes such well records into a national inventory of subsurface lithology and hydraulics (Rafiei, 2026c)). Each record is cured (sentinel and quality flags stripped, physical consistency enforced, units converted), then a variogram is fit and ordinary kriging with k -fold cross-validation produces each surface: static water level, horizontal and vertical hydraulic conductivity, and aquifer thickness. The three-layer column is snapped to the DEM top — an upper and lower drift layer whose combined thickness follows the kriged well-derived aquifer bottom, and a bedrock buffer beneath the kriged water table — so the saturated zone is contained within the active domain everywhere. Storing the well database as a spatially indexed FlatGeobuf reduces a bounding-box query from minutes to under a second.

Recharge. Recharge is taken from the SWAT⁺ model itself — the per-HRU percolation below the soil profile — rather than from a static recharge raster, so it already integrates the weather, soils, land use, and evapotranspiration physics of the calibrated surface-water model. Percolation is mapped to MODFLOW cells with the area-weighted HRU-to-cell map described in Section 2.4 and applied as a long-term annual mean (and, in the coupled mode, as a daily series). A multiplier on recharge is retained as a calibration parameter.

Boundary conditions. The watershed perimeter is a head-dependent general-head boundary (GHB) toward the kriged water table, with conductance scaled by hydraulic conductivity and saturated thickness. This is more defensible than a rigid constant head: it draws near-boundary heads toward the regional water table without prescribing them and without importing the full kriging error of the static-water-level surface. The perimeter is taken directly from the active-domain edge (every active cell with a non-active four-neighbour or on the grid edge, in every layer), which guarantees a complete, leak-free ring rather than the open perimeter cells and interior fixed heads that a buffered-boundary rasterization can leave behind. Lakes are represented as a surface-seepage drain at the land surface. The flow solver uses the Newton formulation with a backtracking, delta-bar-delta under-relaxation configuration tuned for robust convergence at 250 m resolution.

Streams. The stream network — 1,506 streamflow-routing (SFR) reaches built from the SWAT⁺ channel graph, detailed in Section 2.4 — represents only channels of stream order ≥ 2 . Order-1 flowlines are largely artificial headwater traces that sit high in the landscape, dry cells when incised, and do not sustain regional groundwater–stream exchange.

2.4 Two-way coupling

The two models exchange state through the MODFLOW 6 BMI/XMI application programming interface, which exposes the solver’s memory in place and runs the GWF6 flow and GWT6 transport models as a coupled exchange. Surface-water percolation drives groundwater recharge; the groundwater–stream exchange returns to the surface through the MODFLOW 6 streamflow-routing (SFR) package, whose reaches then carry the discharged PFAS in-stream via the streamflow-transport (SFT) package (Section 2.6).

Surface to subsurface: recharge. Because SWAT⁺ HRUs are geo-located polygons, each HRU is intersected with the MODFLOW grid to build an area-weighted HRU-to-cell map (no intermediate disaggregated-HRU step is required, unlike the embedded SWAT–

MODFLOW code). Per-cell recharge is the area-weighted sum of the percolation of the HRUs overlapping that cell, divided by the cell area. The map is a sparse matrix, so a daily recharge field is a single matrix–vector product. In coupled mode, the converged steady model is rewritten as a sequence of daily stress periods (a steady spin-up period followed by transient periods with specific storage and specific yield), and recharge is set at the start of each step by writing in place to the MODFLOW recharge memory pointer through the BMI/XMI interface. On the demonstration watershed all 1,096 daily steps of a three-year run converged, with simulated heads tracking seasonal recharge within ± 0.16 m about the steady baseline.

Subsurface to surface: the groundwater–stream exchange. The groundwater–stream exchange is a native per-reach SFR budget term: the head-dependent leakage between each reach and the cell it occupies, positive when groundwater discharges to the stream (gaining/baseflow) and negative when the stream loses to the aquifer. SFR computes stage from routed flow and so enables in-stream transport, whereas the simpler river (RIV) package prescribes a fixed stage per cell and can route neither water nor solute. Because each reach is already a segment of a SWAT⁺ channel, the exchange aggregates directly to the channel network with no spatial join.

The SFR reach network is built from the SWAT⁺ channel graph. Each channel (filtered to stream order ≥ 2) is oriented upstream-to-downstream by the DEM elevation of its endpoints and intersected with the grid to give one reach per crossed active cell; reaches inherit the channel width, a slope computed from the channel’s elevation drop over its length, a streambed top set below the land surface, and a streambed hydraulic conductivity. Connectivity follows a single-successor graph (each reach has one downstream reach: the next cell within a channel, or the first reach of the receiving channel at a confluence), with reaches numbered in topological order so that every reach’s downstream neighbour has a higher index — matching the solver’s preferred ordering and avoiding a sign ambiguity in

the zero-based connection list. On the demonstration watershed the network has 1,506 reaches across 273 channels with no connectivity violations.

2.5 Groundwater flow calibration with a baseflow constraint

The calibrated flow field is simultaneously head- and flux-consistent: the deployed posterior realization reproduces 5,383 static water-table heads at a Nash–Sutcliffe efficiency of 0.91 (RMSE 5.6 m) while also reproducing the basin’s gaining baseflow, a joint property a head-only fit cannot guarantee. This is achieved by calibrating the groundwater flow model to both water-table elevations and net baseflow with the PEST++ iterative ensemble smoother (White et al., 2020).

The adjustable parameters (Table 2) are a smooth hydraulic-conductivity field of 243 pilot-point log-multipliers interpolated to the grid, supplemented by global multipliers on vertical anisotropy, recharge, drain conductance, the watershed-divide boundary conductance, well pumping, and streambed conductance (the SFR analogue of river conductance). Observations are 5,383 static water-table heads — the per-cell median of the *raw* measured static water levels in the well database (head = top – SWL), not values drawn from the kriged water-table surface — plus a single net-baseflow observation. The targets are therefore independent measurements rather than a smooth product, with one noted partial non-independence: the general-head boundary is set from the kriged surface derived from the same database. We emphasise the RMSE (5.6 m) over the Nash–Sutcliffe efficiency (0.91), since the latter is partly carried by the large topographic range of heads.

The baseflow constraint is essential. A head-only inversion is non-unique and can fit heads while producing an unphysical (net-losing) stream network, so the basin groundwater-to-stream exchange (from the SFR budget) is added as a strongly weighted target equal to the observed gaining baseflow ($5.56 \text{ m}^3 \text{ s}^{-1}$; USGS 04118500 mean flow $7.51 \text{ m}^3 \text{ s}^{-1}$, baseflow index 0.74). Two domain controls keep the constrained problem well posed: raw well *pump capacity* overstates long-term withdrawal in this heavily developed basin and is scaled by

a calibrated multiplier, and the watershed perimeter is treated as a near-no-flow divide (a topographic divide is a groundwater flow divide; a leaky boundary exports baseflow out the sides). The ensemble converges in ten minutes on a single workstation; the posterior realization closest to the observed baseflow gives the head Nash–Sutcliffe efficiency of 0.91 (RMSE 5.6 m over 5,383 heads) and a gaining baseflow of $+5.46 \text{ m}^3 \text{ s}^{-1}$. This flow field is the basis for the groundwater transport.

2.6 Groundwater PFAS transport and integration

PFAS groundwater transport is simulated with a MODFLOW 6 groundwater-transport (GWT) model run on the calibrated flow field through the GWF6–GWT6 exchange, so the transport uses the exact flow solution. The model solves advection (total-variation-diminishing), dispersion ($\alpha_L = 10 \text{ m}$, $\alpha_T = 1 \text{ m}$ at the 250 m grid), and *Freundlich* sorption (MODFLOW 6 MST package; $K_f = 0.005$, exponent $n = 0.8$), the standard PFAS isotherm, giving a concentration-dependent retardation factor. This first implementation simulates PFOS groundwater transport; the multi-analyte source fingerprinting reported in Section 3 is an observation-based decomposition independent of the transport model, and compound-specific groundwater transport with per-analyte sorption isotherms is a priority for future work.

Prescribed source. The PFAS source is set from measurements rather than fitted: there is no calibrated groundwater concentration multiplier. Constant-concentration cells are placed at the source-zone grid cells whose measured groundwater PFOS exceeds 10^4 ng L^{-1} , at their observed concentration. The single peak sample reaches $1.5 \times 10^6 \text{ ng L}^{-1}$; the source is capped at 10^5 ng L^{-1} , because that one peak makes the six-order-of-magnitude front numerically intractable and because the log-space validation tests the plume *shape* rather than the absolute peak (the source magnitude carries the single-sample uncertainty). A small ambient background (10 ng L^{-1} , near the observed median) both removes the Freundlich

C^{n-1} singularity at $C \rightarrow 0$ and compresses the dynamic range; adaptive time stepping carries the stiff transient over a 40-year horizon (the tannery legacy). The transport runs as a transient on the calibrated *steady* flow field, a simplification justified by the multi-decadal source history being long relative to seasonal flow variation but which omits non-stationary pumping and recharge over that period — a limitation for the transient plume timing, though the validation compares time-aggregated plume concentrations. The field is thus a constrained prediction: bounded by the prescribed source, the cap, and the background, but with no fitted concentration term. The groundwater source magnitude and location are therefore fixed by measurement; the only quantity fitted to in-stream observations is the fraction of the discharged load that reaches the stream — the joint-calibration scalar g (Section 3) — not the source itself.

Validation. The modeled groundwater plume is validated against the 846 measured groundwater PFOS observations in the watershed — assigned to grid cells by our extraction pipeline, which snaps the inventory to the MODFLOW 6 grid built in Section 2.6 — excluding the source-zone cells (Table 3). Skill is reported in log space (root-mean-square error, bias, fraction within a factor of ten), the appropriate metric for concentrations spanning many orders of magnitude, and is *stratified by concentration band*: a factor-of-ten miss near the ~ 2 ng L⁻¹ detection limit is measurement noise, whereas the same miss at tens of ng L⁻¹ is a model error, and a single pooled statistic conflates the two.

Porewater discharge validation. A second, more direct test validates the groundwater contribution at the discharge interface itself, rather than only in the far-field plume: streambed *pore water* sampled along the lower mainstem (EGLE North Kent inventory; 43 sites, all within 100 m of a channel and concentrated on the source-bearing reaches) measures the PFAS concentration of groundwater as it enters the stream. The modeled aquifer concentration at each pore-water cell is compared to the measurement in log space by concentration band. This is the most direct available check of the modeled pathway, since pore

water samples the discharging groundwater in place rather than the plume upgradient of it.

Integration with the surface-water pathway. The groundwater-discharged PFAS reaches the stream through the MODFLOW 6 streamflow-transport (SFT) package, which routes it downstream along the SFR channel network. The groundwater PFAS *load* delivered to each reach is the reach gaining flux (the SFR groundwater-exchange budget term, $\text{m}^3 \text{d}^{-1}$) times the aquifer PFAS concentration at that cell (ng L^{-1}), excluding the artificial constant-concentration source cells. Summed over the network and combined with the SWAT⁺ surface streamflow, this groundwater contribution to in-stream PFOS adds to the land-derived load from the surface-water engine, closing the continuous surface-water-plus-groundwater PFAS mass balance.

3 Results

3.1 Surface-water engine performance

Before the groundwater leg is added, the surface-water PFAS engine is validated on its own across the full Rogue network. The implementation passes its correctness and conservation checks: the soil three-phase equilibrium solver reproduces a 40-digit decimal reference to a worst-case relative error of 1.7×10^{-7} over 29,160 parameter combinations, the in-stream routine agrees with the established SWAT⁺ pesticide channel solver (reduced to its linear-partition subset) to a maximum absolute difference of 2.4×10^{-7} kg, and the land- and channel-phase mass balances close to a relative residual of order 10^{-5} . The uncalibrated model, initialized from land-use and soil-class background concentrations alone, reproduces the spatial *pattern* of ambient PFOS — including the mainstem downstream gradient — but overpredicts the *magnitude* by a nearly uniform factor of approximately nine. Because reach concentration is linear in soil loading, a single global soil-loading scale factor of 0.11 corrects the magnitude in one step, reducing the log-space root-mean-square error across the

20 gauged reaches (29 stations) from 1.33 to 0.74 dex and the percent bias from +1516% to +78%. The improvement is concentrated where the model is physically anchored: along the seven source-bearing mainstem reaches the calibrated log-RMSE is 0.15 dex (a factor of 1.4) with a log-space Nash–Sutcliffe efficiency of +0.61 and a percent bias of -7% , and the modeled concentration rises monotonically downstream from 6.2 ng L⁻¹ in the headwaters toward the Wolverine corridor against an observed rise from 6.3 to 27 ng L⁻¹. The calibrated in-stream PFOS field renders this gradient over the full 590-reach network (Figure 3), and the remaining misfit is structured rather than random: the two lowest mainstem reaches underpredict the observed peak (14.7–15.4 vs 19–27 ng L⁻¹), the signature of the concentrated House Street source that a uniform soil background cannot reproduce, while the bidirectional tributary scatter reflects per-land-use heterogeneity in soil PFAS. These per-HRU inputs — the initial soil PFOS pool and the discrete, soil-class-controlled Freundlich sorption fields the engine reads (Figure 4) — are the national, automatable initialization that replaces the measured per-site soil inventory available only for data-rich legacy basins.

3.2 Surface-water PFAS

The surface-water engine reproduces ambient PFOS on the Rogue from the single global soil-loading scale factor calibrated above: along the seven source-bearing mainstem reaches the calibrated log-space root-mean-square error is 0.15 dex (a factor of 1.4), and over all 20 gauged reaches it is 0.74 dex (Table 3). The model captures the mainstem downstream gradient toward the Wolverine corridor (rising from ~ 6 to ~ 27 ng L⁻¹); the all-network residuals are structured rather than random, localizing the concentrated point source and per-land-use soil heterogeneity. This land-derived in-stream PFAS load is this paper’s surface-water-alone result and is the baseline to which the groundwater pathway is added below (Figure 5).

3.3 Groundwater model generation and flow calibration

The automated pipeline generates a complete, calibrated MODFLOW 6 groundwater flow model for the Rogue with no manual editing, yielding a simultaneously head- and flux-consistent field: the pilot-point inversion with a baseflow constraint reaches a head Nash–Sutcliffe efficiency of **0.91** (RMSE 5.6 m over 5,383 static water-table heads) while reproducing the observed gaining baseflow (+5.46 vs 5.56 m³ s⁻¹; Table 3). The model comprises $\approx 24,900$ active cells and a connected SFR network of 1,506 reaches built directly from the SWAT⁺ channel graph. Reaching this on a second, heavily-developed and dissected watershed — beyond the lightly-developed basin on which the generator was first built — requires three reproducible ingredients: scaling raw well pump *capacity* to actual withdrawal, a near-no-flow watershed-divide boundary, and a uniformly-thick bedrock layer for steady-state Newton stability. Two calibrated multipliers carry the weight of these adjustments and should be read as effective, not literal, values (Table 2): the well-pumping multiplier settles near -1.26 in $\log_{10}$ — actual withdrawal $\sim 5\%$ of raw registered capacity, absorbing the gap between permitted capacity and long-term-average use — and the drain conductance rails at its upper prior bound, standing in for unresolved near-surface drainage structure rather than a measured conductance.

3.4 Groundwater PFAS plume, validated against the Wolverine measurements

The Freundlich GWT model reproduces the House Street plume on the calibrated flow field, validated against 846 measured groundwater PFOS observations. The measurements aggregate to 73 grid cells (the per-cell median; “846 observations” is the data volume, not the independent degrees of freedom), of which 8 source-zone cells are *prescribed* and the remaining 65 are *predicted*. Over all 65 predicted cells the log-space RMSE is 1.1 dex with 71% within a factor of ten, and among the 26 predicted cells *above* background (observed

444 $> 50 \text{ ng L}^{-1}$) it is 1.4 dex with 46% (12 of 26) within a factor of ten (Table 3; Figure 6;
 445 Supplementary Figure S5). The source is *prescribed* to the measured plume (constant-
 446 concentration cells set to observed values $> 10^4 \text{ ng L}^{-1}$), so the skill assessment separates
 447 prescribed from predicted cells: the model captures the plume *shape near the source* and
 448 predicts ambient background away from it. The all-cell figure is flattered by the many
 449 background-dominated cells where both model and observations sit near the ambient floor;
 450 the above-background figure is the more demanding test. The elevated observations distant
 451 from House Street indicate additional source areas the single prescribed source does not
 452 represent, not a transport error. At this resolution a grid Péclet number of ~ 25 ($\Delta x / \alpha_L =$
 453 $250/10$) means numerical dispersion dominates the prescribed physical dispersivity. The
 454 streamflow-transport package routes the groundwater-discharged PFAS downstream through
 455 the channel network.

456 A more direct test validates the pathway at the discharge interface itself. Streambed
 457 pore water along the lower mainstem (43 sites, all within 100 m of a channel) measures the
 458 groundwater concentration where it enters the stream, and its skill is sharply concentration-
 459 dependent: below 10 ng L^{-1} the model is essentially unbiased (-0.19 dex, 82% within a
 460 factor of ten), whereas above 50 ng L^{-1} it under-predicts systematically (-1.43 dex). The
 461 deficit is therefore not random and not a near-detection-limit artefact: it is localized to the
 462 highest-concentration discharge reaches near the source, where the 250 m cell smears the
 463 steep plume gradient — the same numerical-dispersion limit identified above, now pinned
 464 to the reaches that carry the bulk of the discharged mass. Reporting a single pooled error
 465 (1.2 dex) would hide this structure and is avoided here.

466 Because the comparison above is in concentration, it understates the mass implication.
 467 The conserved quantity is load (concentration times the gaining flux): the net groundwater-
 468 to-stream exchange integrates to $+5.46 \text{ m}^3 \text{ s}^{-1}$, matching the calibrated baseflow, and the
 469 transport-delivered groundwater PFOS load to the network is of order 45 kg yr^{-1} . Two
 470 reaches of equal concentration but differing gaining flux carry proportionally different mass,

so the validation and the surface/groundwater apportionment are assessed in load space, not concentration alone.

3.5 Detecting the groundwater contribution model-free: multi-analyte source fingerprinting

Before fitting any model to the in-stream data, we test whether the groundwater contribution is detectable in the observations themselves. PFAS compounds of different perfluorocarbon chain length partition differently between the surface and subsurface pathways — short-chain compounds are weakly sorbed and mobile, long-chain compounds are strongly retained — so the legacy groundwater plume and the diffuse surface signal carry distinct chemical fingerprints. Across the watershed the two end-members are sharply separated (Figure 7a): the groundwater plume is PFOS-dominated (PFOS is 0.76 of the five-analyte sum of PFOS, PFOA, PFHxS, PFBS and PFHxA), whereas the in-stream surface signal is short-chain-rich (PFBS and PFHxA together 0.42, versus 0.05 in the plume). Their composition distance is near-maximal ($L_1 = 0.98$ of a possible 2), so an in-stream sample is an identifiable two-end-member mixture.

Decomposing each mainstem in-stream sample into the surface and groundwater end-members by non-negative unmixing — introducing no fitted transport parameter — establishes a clear downstream gradient (Figure 7b): the groundwater fraction rises from ≈ 0 in the headwaters to 0.55–0.63 at the lower mainstem near the Wolverine source. This decomposition is model-free — unlike a model fit it does not rest on the aquifer-side validation, and unlike a single-analyte comparison it cannot be reproduced by any co-located surface source with a different chemical signature. It is thus the primary evidence that the lower-mainstem PFOS is groundwater-borne; the joint calibration below then *quantifies* that contribution and independently recovers the same spatial gradient (Pearson $r = 0.58$).

3.6 Quantifying the groundwater contribution: joint calibration

Having detected the groundwater contribution model-free, we quantify it by jointly fitting the surface and groundwater legs against the same data. The fit credits a statistically significant groundwater term and closes the lower-mainstem residual: the fitted groundwater scalar is $g = 0.061$ (95% CI [0.022, 0.100], excluding zero), significant against the surface-only ($g = 0$) model ($F_{1,5} = 16.2$, $p = 0.01$), and the mainstem log-RMSE falls from 0.15 to 0.07 dex (Figure 8; Table 4). The improvement survives leave-one-out cross-validation across the seven reaches (held-out RMSE $6.2 \rightarrow 4.5$ ng L⁻¹), so it is not a free parameter fitting in-sample. The groundwater contribution rises from $\sim 13\%$ of the predicted concentration in the headwaters to $\sim 53\%$ at the lowest reach near the Wolverine corridor, closing the sharp rise to 19–27 ng L⁻¹ the surface model alone under-predicts (14.7–15.4).

The joint fit is also the correct accounting. A naive sum of the two separately-calibrated legs over-predicts the observed in-stream PFOS by about an order of magnitude because it double-counts: the surface soil-loading factor was fitted to grab samples that already integrate the groundwater contribution. Because in-stream concentration is linear in both the surface soil-loading and the groundwater source strength, the combined prediction at mainstem reach i is $C_i = (L/L_0) C_i^{\text{SW}} + g B_i$, where C_i^{SW} is the surface-water model’s in-stream PFOS at the reference loading $L_0 = 0.11$, B_i is the cumulative groundwater-discharged load routed to reach i divided by its streamflow, and (L, g) are fitted together by non-negative least squares against the observed mainstem gradient in linear concentration space (the dex figures quoted above are the derived diagnostic, not the fitted objective). (The linear superposition assumes in-stream channel processes — settling, resuspension, burial — act linearly and identically on surface- and groundwater-derived PFAS, a defensible first-order approximation given the low ambient sediment load.) Fitting jointly drops the surface soil-loading from $L_0 = 0.11$ to $L = 0.077$ — the signature of resolved double-counting, the groundwater leg now explaining part of the in-stream signal the surface-only calibration had absorbed. The gain is not free: the joint fit slightly under-predicts the headwaters and over-predicts the

first lower-mainstem reach, consistent with a metric weighted toward the high-concentration reaches, and the inference rests on only $n = 7$ source-bearing reaches with $p = 2$ free scalars. The reported significance ($p = 0.01$) is moreover optimistic: consecutive mainstem reaches share cumulative upstream load, so the seven residuals are spatially autocorrelated and the effective sample size is below seven, inflating the nominal degrees of freedom and understating the true p . The groundwater term is in practice carried by the two lowest reaches — the only ones where the groundwater fraction exceeds 50% — so the cross-validation “survives” largely because leaving one of them out still leaves the other as an anchor, and the effective n for the g inference is closer to two than to seven. (Because $g = 0$ lies on the boundary of the non-negativity constraint, the standard F null is also only approximate, an error that is conservative here.) The result remains the most defensible reading of these data, but it is observation-*consistent* support, not a strong significance claim.

Two cautions bound the interpretation. First, g is a single lumped scalar rescaling the modeled groundwater load; it absorbs source-magnitude error, interception, discharge geometry, and the grid-controlled numerical dispersion (grid Péclet ~ 25) together, and is not a clean physical interception fraction. Second, the joint fit establishes that *something* concentrated at the lower mainstem is needed but on concentration alone cannot, by itself, distinguish groundwater discharge from any co-located unmodeled term (an additional surface point source, a tributary input, near-channel resuspension) loading the same reaches; the *attribution* to groundwater is established by the model-free multi-analyte fingerprinting above — which a co-located surface source with a different chemical signature cannot mimic — and is corroborated by the aquifer-side validation. A correct georeferencing of the groundwater discharge to the channel network was essential to expose this signal: an earlier model-grid-local mapping mislocated the discharge to a headwater tributary and produced a spurious null ($g \approx 0$) that the spatially-explicit joint test caught but the aquifer-only validation could not.

Predictive uncertainty. Propagating the joint-calibration parameter uncertainty (the fitted (L, g) and their covariance) through the linear in-stream model gives a 95% prediction interval at each source-bearing mainstem reach (Figure 9). The observed PFOS lies within the predictive band at every reach (log-space $R = 0.95$, log-RMSE 0.07 dex, percent bias -1%), and the band widens toward the lower mainstem where the groundwater contribution — and its uncertainty — is largest. This is the predictive uncertainty conditional on the calibrated model structure; the broader parametric uncertainty across the full coupled-model parameter set is screened by the global sensitivity analysis below.

3.7 Global sensitivity: what controls the coupled prediction

A calibration tunes parameters but does not reveal which of the coupled model’s many parameters actually govern its predictions, nor whether the surface and groundwater compartments interact. We therefore screened the full coupled parameter space with the Morris elementary-effects method: 18 parameters spanning the surface-water PFAS engine (soil loading, the in-stream organic-carbon partition, the Langmuir air–water terms, the percolation partition), the groundwater flow model (recharge and five conductance/conductivity multipliers), and the groundwater transport model (Freundlich K_f and n , the dispersivities, source magnitude, and background), over $r = 20$ trajectories (380 coupled evaluations) on a 64-core cloud instance. Each evaluation runs the entire chain — SWAT⁺ for the land-derived in-stream load and MODFLOW 6 for the groundwater flow-and-transport discharge — and we report the mean absolute elementary effect μ^* (importance) and its spread σ (interaction and nonlinearity) for four quantities of interest (Table 5; Supplementary Figure S16). Thirteen percent of the samples — groundwater parameter combinations with near-stalled flow that fail to converge or exceed the transport time budget — did not complete; we compute the elementary effects step-by-step so that every other evaluation still informs the screening (a median of 17 effects per parameter).

The screening reproduces, from global variance rather than a single fit, the spatial finding

574 the joint calibration reached. At the lower mainstem near the Wolverine source the in-stream
 575 PFOS is controlled by the *groundwater* pathway — the prescribed source magnitude (the
 576 largest μ^*), the aquifer hydraulic conductivity, and the streambed conductance that meters
 577 the discharge — with the surface soil-loading a secondary control; at the mid mainstem
 578 the ranking inverts and the surface soil-loading dominates, the groundwater terms falling
 579 to a few percent of its influence. The groundwater plume concentration is governed by the
 580 transport parameters (source, Freundlich n and K_f) and the baseflow purely by the flow
 581 parameters (recharge, streambed and aquifer conductivity) — exactly the separation the
 582 model structure implies, and a check that the coupling carries each signal to the right place.
 583 The interaction signal is itself diagnostic: for the surface soil-loading $\sigma/\mu^* \approx 0.5$, a near-
 584 linear and additive control, whereas for the groundwater source and conductivity $\sigma/\mu^* \approx 1$,
 585 the strong nonlinearity expected of Freundlich sorption and a threshold-like discharge. This
 586 supports the linear superposition the joint calibration assumed for the surface leg, and it
 587 flags the groundwater source and transport parameters as the ones that warrant variance-
 588 based decomposition. A Sobol analysis on a random-forest surrogate fit to the screening
 589 ensemble (Section 4) confirms it: the lower-mainstem prediction variance is dominated by
 590 streambed conductance, aquifer conductivity and source magnitude (total-order $S_T = 0.53$,
 591 0.41, 0.40), while the mid-mainstem prediction is surface-controlled (soil loading, $S_T = 0.93$)
 592 — the same spatial attribution from global variance.

593 4 Discussion

594 **Groundwater is a plausibly first-order source at the lower mainstem.** The joint
 595 surface-water/groundwater calibration credits a significant groundwater term ($F_{1,5} = 16.2$,
 596 $p = 0.01$), and that term is not diffuse: the groundwater fraction of the in-stream PFOS
 597 rises from $\sim 13\%$ in the headwaters to $\sim 53\%$ at the lowest reach, in the Wolverine corridor,
 598 where it closes the sharp climb to 19–27 ng L⁻¹ that the surface-water model alone under-

Table 1: Study area and coupled-model configuration. The surface-water (SWAT⁺) and groundwater (MODFLOW 6) models are generated automatically from public data on the Rogue River watershed, Michigan.

Quantity	Value
<i>Watershed and observations</i>	
Watershed	Rogue River, Michigan
USGS streamflow gauge	04118500
PFAS contamination source	Wolverine World Wide / House Street site
Source-zone anchor (measured GW PFOS)	cells $> 10^4$ ng L ⁻¹ ; peak 1.5×10^6
Groundwater head observations	5,383 per-cell static water-table heads
Groundwater PFOS observations	846 (downgradient + background)
<i>Surface-water model (SWAT⁺)</i>	
Hydrography source	NHDPlus High Resolution
HRUs	17,771
Channels (SWAT ⁺ graph)	590
<i>Groundwater model (MODFLOW 6)</i>	
Active cells	$\approx 24,900$
Layers	3
Cell size (uniform grid)	250 m
Grid Péclet number	~ 25 ($\Delta x / \alpha_L = 250/10$)
SFR reaches	1,506
SFR channels (order ≥ 2)	273
Porosity	0.30
Bulk density	1800 kg m ⁻³
<i>Transport simulation</i>	
Flow field	calibrated steady state
PFAS transport horizon	40-yr transient (adaptive time stepping)

predicts and lifts the mainstem fit from 0.15 to 0.07 dex (Table 4). Fitting jointly also drops the surface soil-loading ($0.11 \rightarrow 0.077$), consistent with part of the in-stream signal the surface-only calibration had absorbed being reassigned to a groundwater origin. This drop is, however, partly mechanical rather than independent corroboration: because the groundwater load B_i and the surface prediction C_i^{SW} both rise toward the lower mainstem they are positively correlated across the seven reaches, so once g enters as a positive regressor L is algebraically pulled down to hold the overall scale fixed — the elongated, tilted (L, g) ridge of the objective surface (Supplementary Figure S14b). The drop corroborates the partition only to the degree B is more bottom-heavy than C^{SW} (the decorrelation that makes g identifiable at all), a modest margin; the attribution therefore rests on the independent aquifer-side validation, not on the loading drop. The fitted effectiveness $g = 0.061$ is best

Table 2: Calibration parameters. Flow parameters are *calibrated* by the PEST++ iterative ensemble smoother (IES) against 5,383 water-table heads with a baseflow constraint; values are the deployed posterior realization’s global $\log_{10}$ multipliers on the prior field. Transport parameters are *prescribed* from the PFAS literature, and the groundwater source is anchored to measurements rather than fitted.

Parameter	Symbol / units	Prior range	Posterior / used value
<i>Flow (calibrated, PEST++ IES)</i>			
Recharge multiplier	$\log_{10}$ mult.	$[-0.30, 0.48]$	+0.188
Vertical conductivity (anisotropy)	k_v , $\log_{10}$ mult.	$[-0.8, 0.8]$	+0.023
GHB (divide) conductance	$\log_{10}$ mult.	$[-2.0, 0.5]$	-1.169
Drain conductance	$\log_{10}$ mult.	$[-2.0, 0.5]$	+0.500 ^a
Well pumping / withdrawal	$\log_{10}$ mult.	$[-1.70, 0.0]$	-1.264
SFR streambed conductance	$\log_{10}$ mult.	$[-1.5, 1.0]$	+0.693
Hydraulic conductivity field	243 pilot points, $\log_{10}$ mult.	$[-0.8, 0.8]$	spatially variable
<i>Transport (prescribed, literature)</i>			
Freundlich sorption coefficient	K_f	— ^b	0.005
Freundlich exponent	n	— ^b	0.8
Longitudinal dispersivity	α_L (m)	— ^b	10
Transverse dispersivity	α_T (m)	— ^b	1
Molecular diffusion	D (m ² s ⁻¹)	— ^b	1×10^{-10}
Source cap	ng L ⁻¹	— ^b	1×10^5
Source threshold	ng L ⁻¹	— ^b	1×10^4
Ambient background	ng L ⁻¹	— ^b	10

^aThe drain-conductance multiplier reaches its upper prior bound (+0.5), indicating the calibration favours stronger drainage than the prior centre; the remaining flow multipliers are interior. ^bPrescribed PFAS-transport literature value; not calibrated, so no prior range applies.

610 read as a calibrated statement that only $\sim 6\%$ of the modeled groundwater PFAS load reaches
611 the stream, the remainder lost to source over-representation, discharge geometry the 250 m
612 grid cannot resolve, and grid-controlled numerical dispersion — and *not* to the engineered
613 Rogue/Rum Creek interceptor, which post-dates the 2019 validation samples by six years.
614 The magnitude is poorly constrained — seven mainstem reaches, and the linear joint fit
615 cannot by itself separate groundwater discharge from any unmodeled term co-located near
616 the source. The attribution to groundwater therefore rests on the independent aquifer-
617 side validation: the transport model, with its source prescribed (not fitted) to the measured
618 plume, reproduces the near-source PFOS shape against 846 groundwater observations before
619 any in-stream fitting (Table 3), to $\sim 1.1 \log_{10}$ units (dex) with 71% of predicted cells within
620 a factor of ten of the measurement. That level of agreement is the accepted bar for regional
621 plume models, where routine measurement and sampling error alone propagate to order-of-

Table 3: Consolidated validation metrics for the coupled system across the groundwater-flow, groundwater-transport, surface-water-transport, and joint compartments. Concentration skill is reported in $\log_{10}$ space (dex), the appropriate metric for values spanning orders of magnitude.

Compartment / quantity	Metric	Value	n	Method / notes
Groundwater heads	Nash–Sutcliffe / RMSE	0.91 / 5.6 m	5,383	PEST++ IES, baseflow-constrained
Baseflow	gaining discharge	+5.46 vs 5.56	—	$\text{m}^3 \text{ s}^{-1}$; observed gaining baseflow
Groundwater PFOS (all predicted cells)	log-RMSE / within $\times 10$	1.1 dex / 71%	65	846 obs $\rightarrow$ 73 cells; 8 prescribed, 65 predicted
Groundwater PFOS (above background)	log-RMSE / within $\times 10$	1.4 dex / 46%	26	observed $> 50 \text{ ng L}^{-1}$; 12/26 within $\times 10$
Surface-water PFOS (source-bearing mainstem)	log-RMSE	0.15 dex	7	calibrated mainstem reaches
Surface-water PFOS (all gauged reaches)	log-RMSE	0.74 dex	20	negative log Nash–Sutcliffe; tributary heterogeneity
Joint SW+GW mainstem fit	log-RMSE; held-out RMSE	0.07 dex; $6.2 \rightarrow 4.5$	7	NNLS (L, g); held-out RMSE in ng L^{-1} (Fig. S14 reports MAE)

magnitude scatter in plume quantities (Jarsjö et al., 2008); the factor-of-2 fits reported in the literature are achieved only at heavily instrumented single-source sites with dense, dedicated monitoring networks (Shin et al., 2011), which the present automated, public-data pipeline does not assume.

Why the groundwater source is legacy rather than present-day recharge. Treating the groundwater source as a prescribed, measurement-based plume (Section 2.6) rather than a flux from current land loading is justified by a vadose-zone travel-time analysis. Applying the model’s own three-phase retardation — Freundlich solid sorption plus a Langmuir air–water-interface term — over the kriged depth-to-water (median 31 m) gives a unsaturated-zone transit of *centuries* for PFOS (median $\sim 10^3$ yr; 5–95% range ~ 440 –5,500 yr) and *decades-to-centuries* for the shorter-chain PFOA (~ 220 yr median). The

Table 4: Joint surface-water + groundwater calibration along the Rogue mainstem (upstream $\rightarrow$ downstream). In-stream PFOS is linear in both the surface soil-loading L and the groundwater source effectiveness g ; fitting (L, g) jointly by non-negative least squares gives $L = 0.077$ (from a surface-only $L_0 = 0.11$) and $g = 0.061$ (95% CI $[0.022, 0.100]$). The added groundwater term is significant against the surface-only ($g = 0$) model ($F_{1,5} = 16.2$, $p = 0.01$) and survives leave-one-out cross-validation, and the groundwater fraction rises from $\sim 13\%$ in the headwaters to $\sim 53\%$ at the lower mainstem near the Wolverine source. Concentrations in ng L^{-1} .

Channel	Observed	SW-only	Joint	SW part	GW part	GW frac. (%)
26	6.3	6.2	5.0	4.3	0.6	13
18	6.8	7.7	6.1	5.4	0.7	11
15	7.2	9.2	7.2	6.4	0.7	10
11	6.7	11.2	8.6	7.8	0.8	9
10	9.6	12.4	9.5	8.7	0.8	8
1	19.0	14.7	23.0	10.3	12.7	55
2	27.0	15.4	22.8	10.8	12.0	53

~ 4.5 -fold PFOA-precedes-PFOS contrast follows from the carboxylate’s lower solid- and air–water-interface partitioning — the same chain-length ordering reported for distributed watershed PFAS transport (Rafiei and Nejadhashemi, 2023). A lumped travel-time estimate and an explicit one-dimensional advection–dispersion–retardation solution return identical 50% breakthrough times, and a variance-based sensitivity ranks recharge, vadose thickness, and grain size (through the air–water interfacial area) above the partition coefficients themselves. The air–water interface alone adds a $1.3\text{--}5\times$ retardation that strengthens as the medium fines, within the range reported for unsaturated PFAS transport. Because every transit time is far longer than the simulated period, diffuse land-applied PFAS has not yet reached the water table: the observed plume is the historical point source (the tannery legacy), which the coupling routes to the stream network. This is also why a deep explicit unsaturated-zone (UZF/UZT) treatment, while the natural refinement, would not change the legacy-versus-present-day attribution over the simulated window — although, as set out below, it would replace the prescribed source with a mechanistically derived one and is the priority next step.

Table 5: Coupled Morris elementary-effects screening: relative parameter importance (μ^* as % of the most influential parameter for each quantity of interest). Shown are the union of the six most influential parameters per quantity of interest (13 of the 18 screened parameters; the remaining five rank outside the top six for every quantity of interest). Leg: SW = surface-water PFAS, GWf = groundwater flow, GWt = groundwater transport.

Parameter	Leg	In-stream (low)	In-stream (mid)	GW plume	Baseflow
source	GWt	100	2	100	0
K_h	GWf	69	3	1	35
soil load	SW	60	100	0	0
SFR-bed K	GWf	54	12	0	62
aq. K_f	GWt	20	2	1	0
Langmuir k_L	SW	17	25	0	0
Langmuir Γ_{max}	SW	9	15	0	0
recharge	GWf	11	9	1	100
background	GWt	3	8	0	0
aq. n	GWt	10	2	2	0
GHB cond.	GWf	4	1	0	7
drain cond.	GWf	2	2	0	25
pumping	GWf	0	0	0	3

Why coupling through the MODFLOW 6 interface matters. The groundwater PFAS pathway, and its routing back to the streams, exists *at all* only because the coupling is built on the current USGS framework through its model interface: multi-model simulation, the GWT transport model with a native Freundlich isotherm, and SFT in-stream transport come for free, without maintaining a forked solver. The same coupling makes the flow inversion well-posed. A head-only inversion is non-unique; adding the groundwater-to-stream exchange — a flux observable only because the coupling computes it — pins down a model that is simultaneously head-consistent (NSE 0.91 over 5,383 heads) and flux-consistent with the observed gaining baseflow.

Automation from public data makes this reproducible at archive scale. Every input is inherited from the already-automated surface-water model — the shared DEM as land surface, recharge from the land-phase water balance, the channel network, aquifer properties kriged from a raw well database — so the coupled model carries no bespoke, hand-edited groundwater build. Reviews of watershed PFAS modeling and the antecedent distributed surface-water model (Rafiei and Nejadhashemi, 2023) identified the missing con-

663 tinuous land–stream–aquifer mass balance as the principal gap; closing it inside an openly
 664 supported transport framework, rather than a frozen one-way reactive-transport coupling,
 665 is what lets the same pipeline run across the national archive of gauged basins, where cou-
 666 pled groundwater contaminant models are otherwise one-off and rarely reproducible. The
 667 integrity of this automation is itself a result: the joint groundwater term is only credible
 668 because it survived a spatially explicit test — an earlier coordinate-referencing error mis-
 669 located the discharge and produced a spurious null that the aquifer-only validation and a
 670 naive additive sum both masked, a failure mode a careful coupled integration is precisely
 671 what surfaces.

672 **Bounds and next steps.** The result stands on one watershed at 250 m, a three-layer
 673 column, a single prescribed source with Freundlich sorption, and a single lumped effectiveness
 674 g ; the validation residuals point to additional source areas the model does not yet include.
 675 Matching *absolute* far-field concentration and mass discharge generally requires processes this
 676 screening model deliberately omits — matrix diffusion and legacy back-diffusion from low-
 677 permeability zones (Kulkarni et al., 2022), and air–water–interfacial-area uncertainty that can
 678 shift predicted groundwater arrival several-fold (Brusseau and Guo, 2021). A trend-captured
 679 but magnitude-under-predicted outcome is therefore expected, and matches our own prior
 680 watershed PFAS result, where PFOS trends were reproduced but PFOA magnitude was
 681 under-predicted for want of diffuse and historical sources (Rafiei and Nejadhashemi, 2023).
 682 Two bounds are numerical: at a grid Péclet of ~ 25 the in-stream estimate carries grid-
 683 controlled numerical dispersion, and the 250 m cell cannot separate the source from the
 684 stream it discharges to. The criterion for controlling numerical dispersion is a grid Péclet
 685 number of order unity ($Pe \lesssim 1-2$), above which a coarse cell injects an artificial dispersivity
 686 of order $\Delta x/2$ (Zheng and Bennett, 2002; Anderson et al., 2015); at $Pe \sim 25$ our grid is
 687 firmly in this numerical-dispersion-dominated regime. Crucially, this spurious mixing can
 688 *mask* a real misfit rather than merely blur it, so a good fit on a coarse advective grid is not

by itself proof of a correct transport model (Benson et al., 2017). The obvious remedy — *local grid refinement* (a quadtree-refined unstructured DISV grid in MODFLOW 6) through the Wolverine corridor (Figure 10) — we tested directly. Refining the source-to-discharge corridor from 250 to 62.5 m did *not* reduce the streambed-porewater under-prediction but slightly widened it (the $> 50 \text{ ng L}^{-1}$ bias moving from -1.46 to -1.86 dex across the analytes that converged), confirming that the coarse-grid numerical dispersion was *masking* a real under-delivery of PFAS to the discharge cells rather than causing it (Benson et al., 2017). The residual discharge-zone error is therefore *structural* — the prescribed single source, the steady flow field, and the sub-grid focused discharge described below — not a discretization artifact, and $g = 0.061$ accordingly remains a lumped scalar rather than a physical interception fraction. (The most strongly sorbing analyte, PFOS, would not converge on the refined grid at all, a further indication that geometric refinement alone is not the limiting lever here.)

A related, partly physical bound shows up as the stratified under-prediction in the high-concentration band of the streambed porewater discharge cells: groundwater upwelling is fragmented by hyporheic flow into sub-metre “pinhole” stream tubes (95% of upwelling areas $< 1.25 \text{ m}^2$; Mojarrad et al., 2023), so a block-averaged 250 m discharge cell dilutes the focused peak concentration by construction, and the PFAS-specific interface processes documented at this very site — organic-matter and air-water-interface retention as the House Street plume discharges to the Rogue River (Lemke et al., 2024) — further complicate the porewater signal. Resolving this would benefit from the streambed-versus-stream-water flux cross-validation approach of Sun et al. (2021). A coupled Morris screening across all eighteen surface- and groundwater parameters (Section 3; Table 5) already establishes which parameters govern the prediction — the groundwater source and aquifer conductivity at the lower mainstem, the surface soil-loading mid-basin — and flags the strongly interacting groundwater source and transport terms. A variance-based Sobol decomposition, computed on a random-forest surrogate fit to the screening ensemble, confirms this split quantitatively: the lower-mainstem in-stream PFOS variance is dominated by the groundwater pathway —

streambed conductance ($S_T = 0.53$), aquifer conductivity (0.41) and source magnitude (0.40) — whereas the mid-mainstem prediction is almost entirely surface-controlled (soil PFAS loading, $S_T = 0.93$). The variance decomposition thus reproduces, from global parameter sensitivity, the same groundwater-lower / surface-mid attribution that the chemistry and the joint fit independently give. The steady flow field is adequate for the time-aggregated concentration match validated here, but not for plume arrival times — the transient extension addressed below. The regulatory thresholds that motivate this work (4 ng L⁻¹ for PFOS) sit well below the present error bars — a factor of ten in groundwater concentration, an order of magnitude in the integrated in-stream estimate — so the model in its current form is a tool for *source attribution and screening*: it identifies which pathways and source areas matter, not whether a numeric limit is met. Closing the gap to decision-grade precision is the job of the joint calibration and the grid refinement above. What we stand behind now is a coupled, observation-validated surface-water/groundwater PFAS model that resolves *which* pathways and source areas govern the in-stream signal — the first to carry the contaminant through both phases and test each against the site’s rich public record.

The priority next step: an explicit vadose zone. The single most consequential extension is to resolve the unsaturated zone explicitly. In the present model the groundwater source is *prescribed* from measurements and the vadose zone enters only as a recharge water flux and the travel-time argument above; yet for PFAS the unsaturated zone is the controlling compartment, because the air–water interface (AWI) that retards transport and stores the largest fraction of the contaminant mass exists only above the water table (Costanza et al., 2019; Brusseau, 2018; Guo et al., 2020). An explicit unsaturated flow-and-transport treatment (MODFLOW 6 UZF coupled to UZT), built on the same kriged depth-to-water that already defines the deep (~ 31 m median) vadose column here, would *derive* the groundwater source from the historical land-loading record rather than prescribing it — converting the lumped delivery factor g into a physically interpretable flux and turning the steady

screening model into a transient one able to address arrival timing and the forward question of whether and when a regulatory threshold is met. The rigorous form of this extension must carry the AWI contribution into the unsaturated transport: standard UZT represents linear or Freundlich solid-phase sorption but not the AWI Langmuir term that dominates PFAS retardation in unsaturated media, so a naive implementation would under-retard and overstate arrival. We expect this extension to *confirm and mechanize* the legacy-source attribution established here — the observed plume would remain the historical point source — rather than overturn it, while substantially raising the model’s predictive reach. Alongside it, generalizing the automated pipeline across a population of gauged basins, extending in-stream transport to the full multi-analyte suite, and constraining the plume’s exposure history with groundwater age tracers (tritium, radiocarbon) at the Wolverine site are the natural complements.

5 Conclusions

We have ported and verified the process-based PFAS surface-water fate-and-transport model of Rafiei and Nejadhashemi (2023) — developed for legacy SWAT — into modern SWAT⁺, where no PFAS capability existed: a soil three-phase equilibrium solver (aqueous, Freundlich solid, and Langmuir air–water interface), land-phase mobilization through runoff, lateral flow, leaching, and sediment, and an in-stream routine. The port is anchored to two field-independent correctness checks: a 1.7×10^{-7} agreement of the equilibrium solver with a 40-digit reference, and numerical identity of the in-stream routine with the established SWAT⁺ pesticide channel solver on its linear-partition subset, so the ported physics is provably confined to the soil column. An automated, public-data framework then couples this engine to a MODFLOW 6 groundwater flow-and-transport model through the first two-way SWAT⁺/MODFLOW 6 coupling, carrying PFAS through a continuous surface-water and groundwater watershed mass balance — to our knowledge the first coupling to carry PFAS

in both directions between a watershed’s surface water and groundwater, with observation
validation in each compartment, advancing the one-way SWAT–MODFLOW–RT3D of Rafiei
and Nejadhashemi (2023). On the Rogue River, the groundwater model — generated on the
fly from public data (SWAT+ DEM as land surface, aquifer properties kriged from a raw well
database, recharge from the water balance) — matches 5,383 water-table observations at a
head Nash–Sutcliffe efficiency of 0.91 under a baseflow constraint while reproducing the ob-
served gaining baseflow. A Freundlich transport model, its source *prescribed* to the measured
Wolverine/House Street plume rather than fitted, reproduces the groundwater PFOS field at
1.1 dex across 65 predicted cells (from 846 observations; 71% within a factor of ten), and SFT
routes the discharged PFAS through the channel network; the surface-water engine repro-
duces the source-bearing mainstem PFOS gradient at 0.15 dex. The groundwater-to-stream
pathway is established first by model-free observational evidence — measured streambed
pore water at the discharge interface carries the source plume’s multi-analyte fingerprint —
and corroborated by a *joint* calibration that fits the surface and groundwater scale factors
together, avoiding the double-counting a naive additive sum incurs. That joint fit (effec-
tive $n \approx 2$) is observation-consistent with a plausibly first-order groundwater contribution
concentrated at the lower mainstem near the Wolverine source, while *reducing* the surface
soil-loading — the signature of resolved double-counting; its magnitude is poorly constrained
at seven reaches, and the attribution rests on the independent aquifer-side and fingerprint
evidence rather than on the calibration alone. By reproducing established loose-coupling
physics inside the open MODFLOW 6 transport framework, anchoring the contaminant
pathway to measurements, automating the build from public data, and integrating the two
compartments through a joint calibration that survives a spatially explicit test, the frame-
work delivers a continuous surface-water/groundwater PFAS mass balance reproducible at
the scale of a national archive. Sharpening the groundwater effectiveness toward a physical
interception fraction through local grid refinement at the source, and generalizing across
basins, are the natural next steps.

Code and data availability

All models, code, and data required to reproduce the results reported here are archived in a public Zenodo repository (DOI 10.5281/zenodo.20838389): the SWAT⁺ PFAS surface-water model, the on-the-fly MODFLOW 6 groundwater flow-and-transport model, the GWF–GWT coupling and SFT in-stream routing, the PEST++ pilot-point calibration with the baseflow constraint, the calibrated flow field for the Rogue River model (USGS 04118500), and the analysis and figure-generating scripts. The SWAT⁺ engine itself — including its PFAS fate-and-transport implementation and the MODFLOW 6 coupling — is developed openly at <https://github.com/rafiei-vahid/swatplus>; its acceleration and byte-reproducibility methodology are documented separately (Rafiei, 2026a).

All inputs are public: USGS streamflow records (U.S. Geological Survey, 2024a) and NHDPlus High Resolution (U.S. Geological Survey, 2024b) for delineation, the groundwater-well database used for the hydrogeologic layers (harmonized at national scale in a companion inventory, Rafiei, 2026c), and the EPA ECHO (U.S. Environmental Protection Agency, 2024a) and Michigan EGLE (Michigan Department of Environment, Great Lakes, and Energy (EGLE), 2024) PFAS source and observation inventories, alongside the MODFLOW 6 and PEST++ executables.

The SWAT⁺ and MODFLOW 6 models analysed here were generated and calibrated with SWATGenX (Rafiei, 2026b) (<https://swatgenx.com>), which automatically builds and calibrates such models from public data for any U.S. Geological Survey gauge in the contiguous United States. SWATGenX is the source of the models and the environment in which this study was carried out; the Zenodo archive is the self-contained package for reproducing the specific results presented here.

CRedit authorship contribution statement

Vahid Rafei: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used a large language model (Anthropic Claude) to assist with code development, analysis scripting, and editing of the manuscript text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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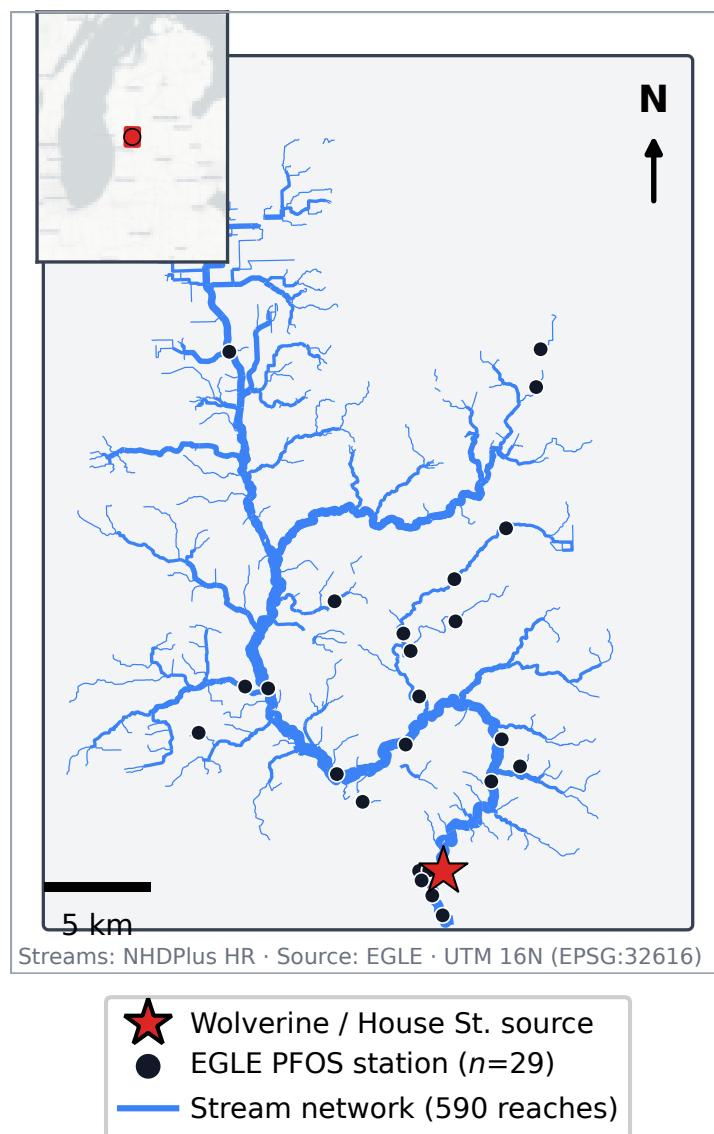


Figure 2: The Rogue River watershed (USGS 04118500), Michigan. The SWATGenX-generated (Rafiei, 2026b) SWAT⁺ network resolves 590 channels from NHDPlus High Resolution; the red star marks the Wolverine World Wide / House Street legacy PFAS source in the lower river. The same watershed carries the dense surface-water and groundwater PFAS observation networks used to validate both compartments. Map lines delineate study areas and do not necessarily depict accepted national boundaries.

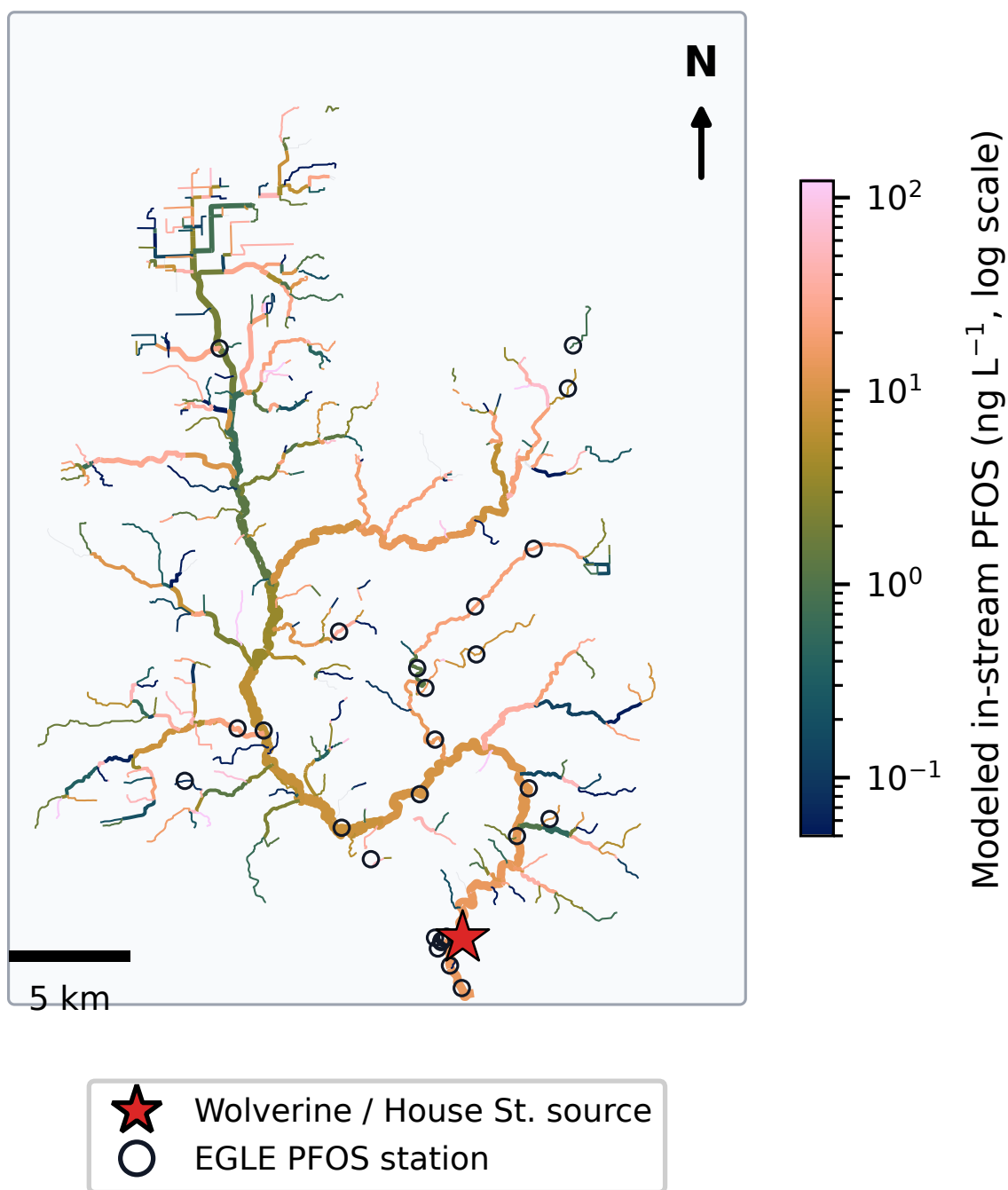


Figure 3: Calibrated (0.11 soil-loading scale, no point source) flow-weighted in-stream PFOS over the 590-reach Rogue network on a logarithmic colour scale. Concentration rises along the mainstem from green tributaries through the orange mid-river toward the Wolverine source (red star); open rings are the EGLE ambient monitoring stations.

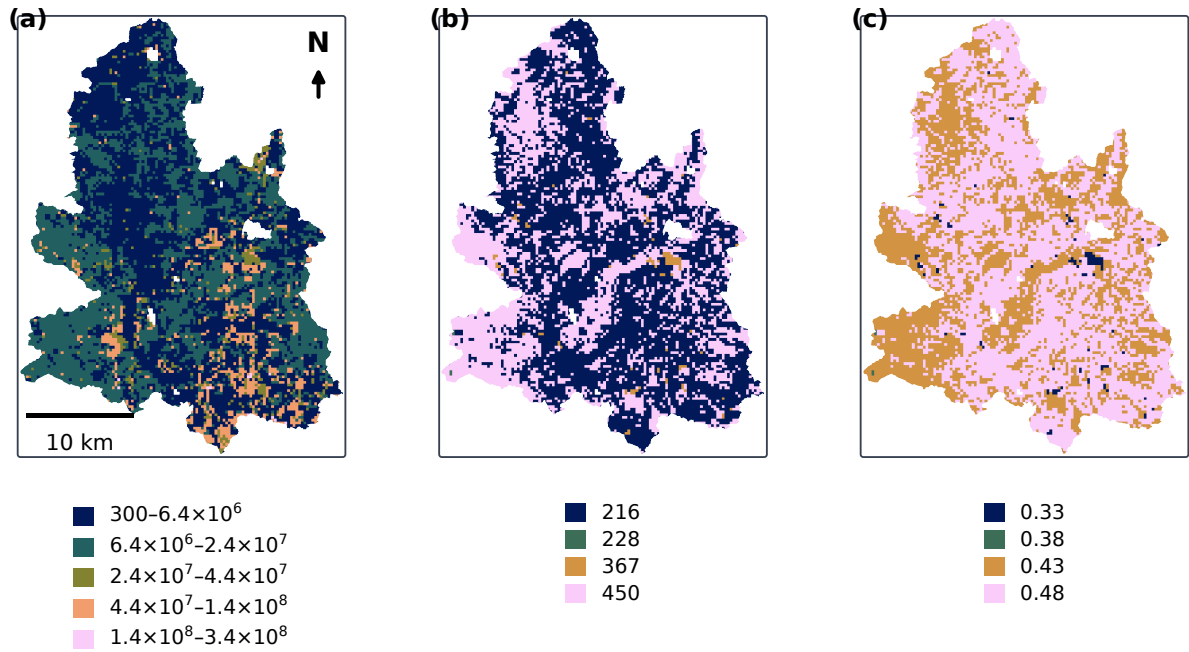


Figure 4: Per-HRU soil-PFAS initialization across the 17,771 Rogue HRUs: (a) initial surface-layer soil PFOS pool (continuous, NaturalBreaks classes); (b) Freundlich sorption coefficient k_f and (c) Freundlich exponent n , both discrete soil-class fields. Colour map: batlow (perceptually uniform, colour-blind safe). The three panels share the same spatial extent; the scale bar and north arrow on panel (a) apply to all.

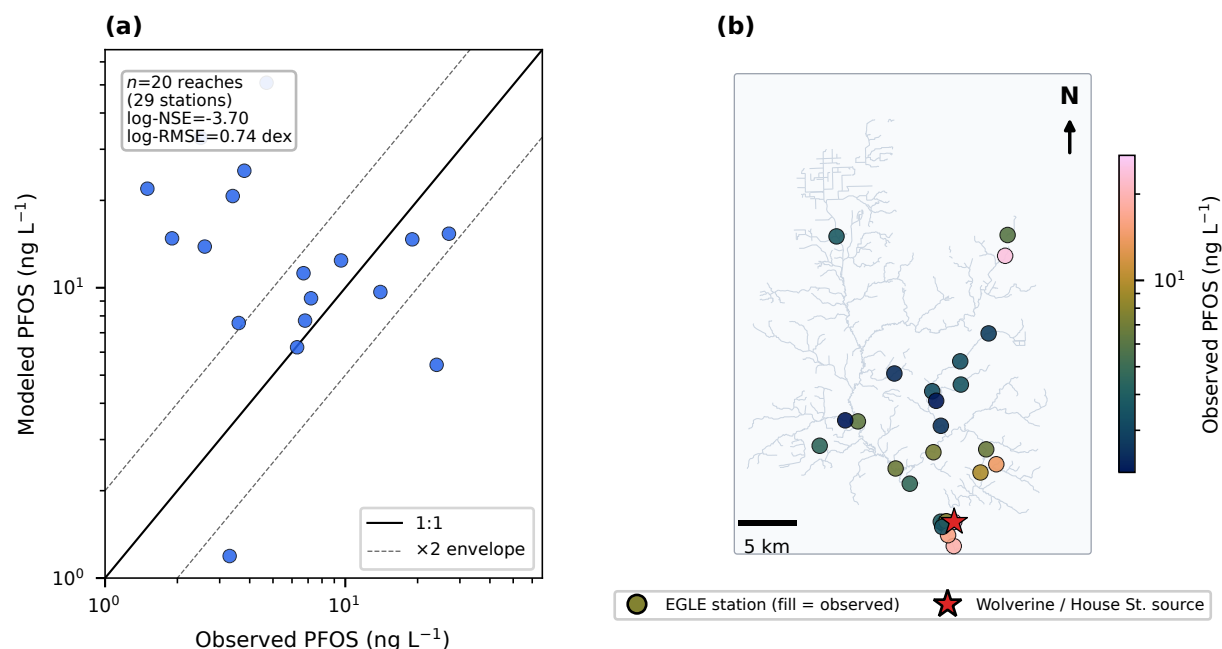


Figure 5: Surface-water PFAS on the Rogue. (a) Observed versus modeled in-stream PFOS, one point per gauged reach; the source-bearing mainstem reaches lie within the $\times 2$ envelope while tributaries scatter (the structured residual that localizes source and soil heterogeneity). (b) The ambient monitoring stations coloured by observed PFOS, showing the downstream rise toward the Wolverine source.

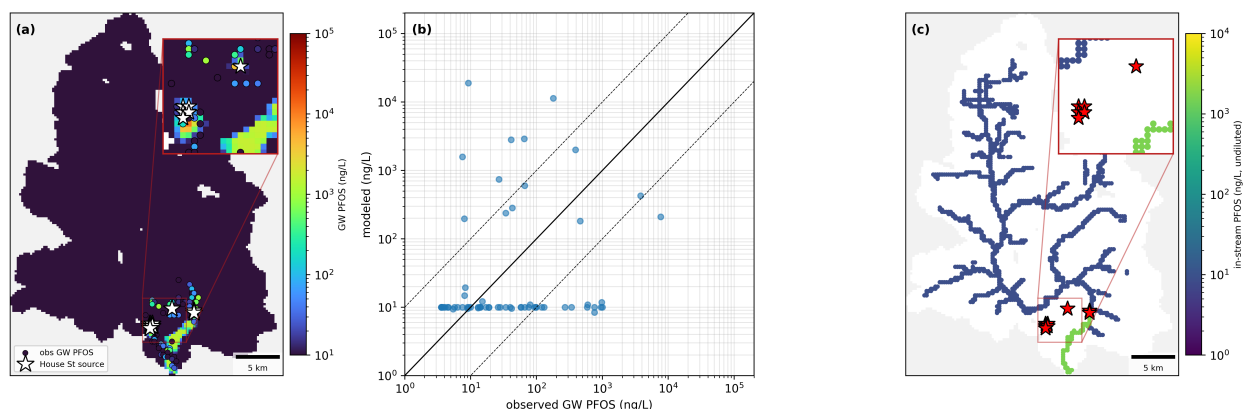


Figure 6: Groundwater PFAS on the Rogue. (a) Modeled groundwater PFOS plume with the 846 measured observations overlaid (same colour scale) and the House Street source; (b) modeled versus observed groundwater PFOS, one point per validation cell, with the 1:1 line and $\times 10$ envelope (panel-title $\log\text{-RMSE} \approx 1.1$ dex over all predicted cells, rounded; the honest above-background figure is 1.4 dex, Section 2.6 and Supplementary Figure S5); (c) the groundwater-discharged PFOS routed by SFT into 1,354 of the 1,506 stream reaches (raw, before dilution by surface streamflow).

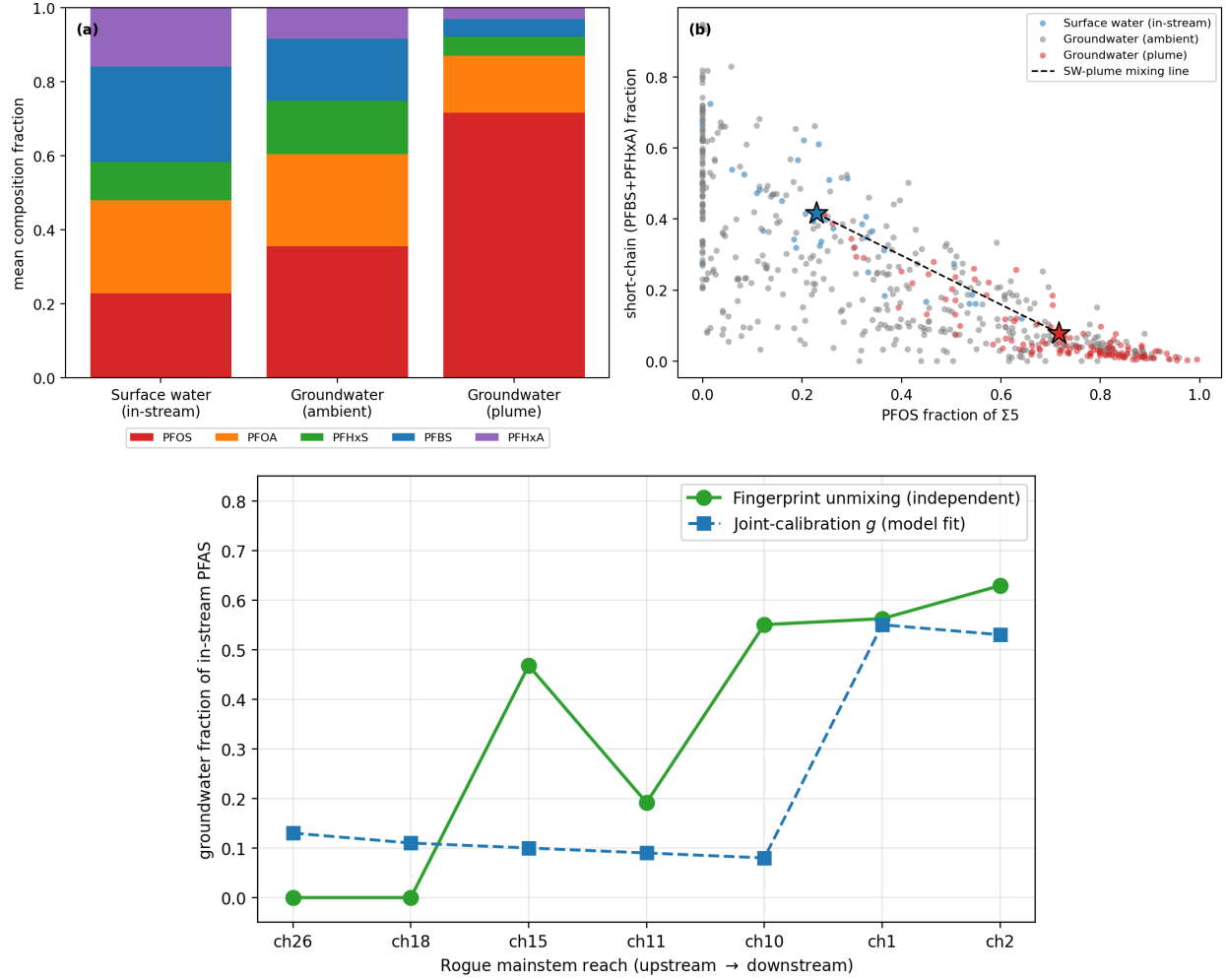


Figure 7: Multi-analyte source fingerprinting. (a) Mean five-analyte composition of the groundwater plume, ambient groundwater, and in-stream surface water; the plume is PFOS-dominated and the surface signal short-chain-rich. (b) Two-end-member mixing space (PFOS fraction versus short-chain fraction); in-stream samples lie on the surface–plume mixing line. (c) Groundwater fraction of in-stream PFAS along the Rogue mainstem from the parameter-free fingerprint decomposition (green) versus the joint-calibration scalar g (blue); the two agree without any shared parameter.

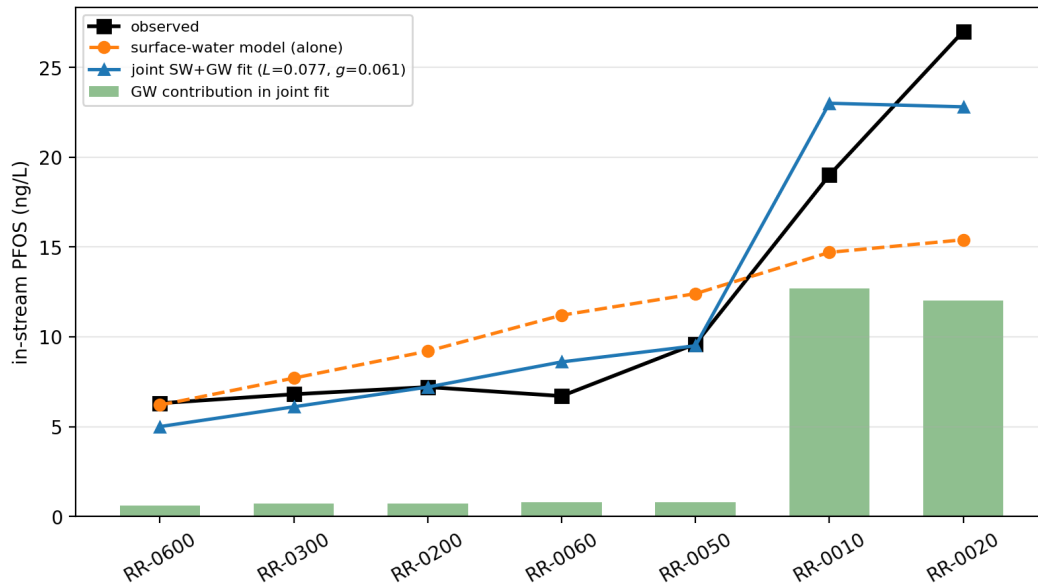


Figure 8: Joint surface-water + groundwater calibration on the Rogue mainstem (upstream to downstream). The joint fit (blue) tracks the observed rise (black) where the surface-water model alone (orange) under-predicts; the fitted groundwater contribution (green bars) is small in the headwaters and large at the lower mainstem near the Wolverine source, closing the lower-mainstem residual ($g = 0.061$; joint 0.07 vs surface-only 0.15 dex).

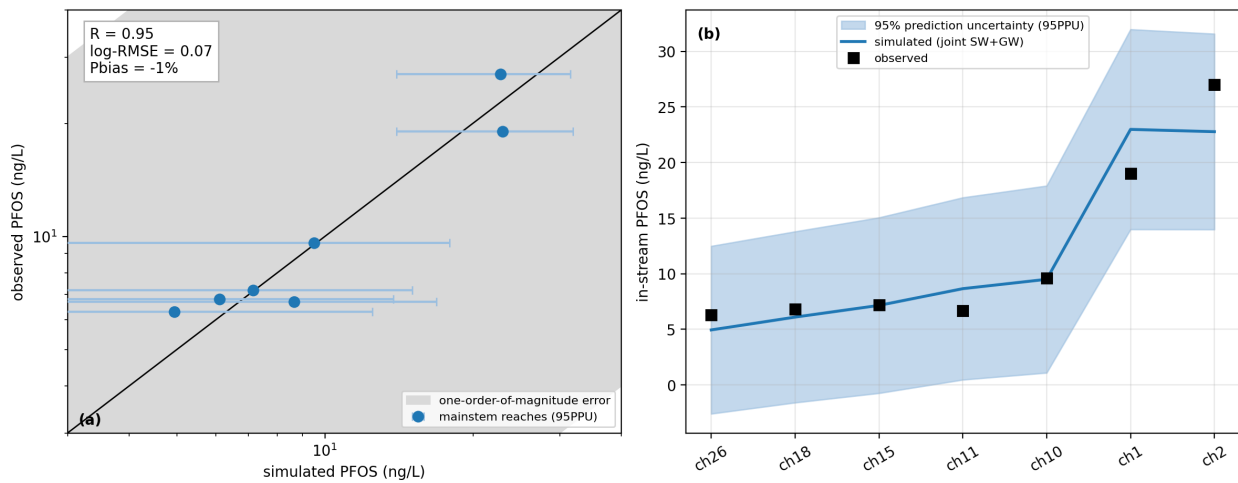


Figure 9: Predictive uncertainty of the joint surface-water + groundwater in-stream PFOS model on the seven source-bearing mainstem reaches. (a) Observed versus simulated PFOS (log axes) with the 1:1 line, the one-order-of-magnitude envelope, and the 95% prediction interval (horizontal bars) propagated from the joint-calibration parameter uncertainty; $R = 0.95$, log-RMSE 0.07 dex, percent bias -1% . (b) The same reaches upstream to downstream, with the 95% prediction band, the simulated profile, and the observations; the band widens toward the groundwater-dominated lower mainstem.

(a)



(b)

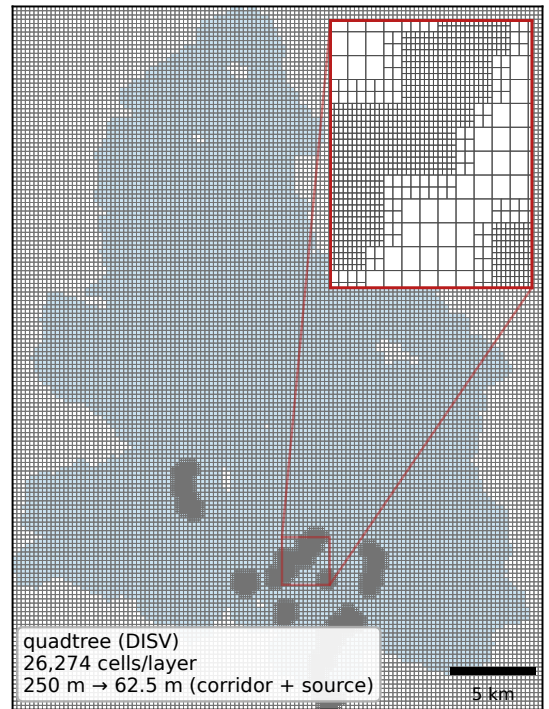


Figure 10: Structured versus quadtree (DISV) groundwater grid used in the refinement test. (a) The uniform 250 m structured grid (11,727 active cells per layer); (b) the quadtree-refined DISV grid (26,274 cells per layer), refined to 62.5 m through the source-to-discharge corridor, with the nested refinement at the Wolverine source shown in the inset. Refining this corridor did not close the streambed-porewater under-prediction, indicating that the coarse-grid numerical dispersion was masking rather than causing it.